AA-O-001628(GSA-FSS)
September 1, 1971

INTERIM FEDERAL SPECIFICATION

CHAIRS, WOOD, GENERAL OFFICE

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C.,20406 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers straight and rotary wood upholstered chairs for general office use.

1.2 Classification.

1.2.1 Types and styles. The chairs shall be of the following types and styles as specified (see 6.2):

Type I. Straight

Style A - Without arms

Style B - With arms

Type II. Rotary

Style A - Without arms

Style B - With arms

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
L-P-00386	- Plastic Material, Cellular Urethane (Flexible).
V-T-276	- Thread, Cotton.
V-T-285	- Thread, Polyester.
V-T-295	- Thread, Nylon.
FF-C-82	- Casters, Swivel, Chair.
FF-N-103	- Nails (Small) and Tacks: Cut.
FF-S-111	- Screw, Wood.
TT-L-57	- Lacquer, Rubbing, Clear, (For Wood Furniture).
PPP-B-00636	- Boxes, Shipping, Fiberboard.
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

FSC 7110

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Commercial Standards:

CS35 - Hardwood Plywood

(Copies of the Commercial Standards referenced above, which are Department of Commerce publications, may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Prices may be obtained from the Superintendent of Documents.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

ASTM D 905-49 - Standard Method of Test for Strength Properties of Adhesion in Shear by Compression Loading

(Copies may be obtained upon application to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished item as well as a cut-away showing the upholstery method and each component part conforming to the requirements of this specification shall be offered for approval of the contracting officer or his authorized representative. Approval of the preproduction sample authorizes the commencement of production but does not relieve the contractor of responsibility from compliance with all other applicable provisions of this specification.

3.2 Materials. The materials used in construction of the chairs shall be of the kind and quality specified herein.

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3.2.1 Adhesives. The adhesives to be used and their permitted application shall be as shown in Table I.

Table I. Adhesives

Application	Kind of Adhesive
All glue blocks, assembly joints, glued laminated stock, and edge joining of lumber-core panels.	Any adhesive that will develop an average block shear strength of not less than 2800 pounds per square inch when tested as specified in 4.3.1.1.
All plywood gluing, edge-joining of veneer plies, cross-banding, and edge banding.	Any adhesive that will conform to or exceed the test requirements of the Type II bond, as described in the Commercial Standard CS35 for Hardwood Plywood.

3.2.2 Wood. The permissible woods for component parts shall be as shown in Table II.

Table II. Wood

Exposed	Unexposed
Beech	Any wood listed under Exposed
Birch	Ash
Cherry	Hackberry
Hickory	Magnolia
Mahogany 1/	Sweetgum
Maple	Sycamore
Pecan	Tupelo
Rockelm	
Walnut	
Philippine hardwood 2/	
White Oak	

1/ American and African Mahogany only.
2/ Red luan group only.

Unexposed and exposed component parts shall be constructed of either one piece of wood or from full-length glued-up stock. There shall be no open joints in the completed component parts. Tests specified in 4.4.2 shall be performed dailu on a minimum of 5 representative glued-up component parts. Glue joints shall show at least 85 percent wood failure.

3.2.2.1 Characteristics and seasoning. The exposed wood parts used for the chairs shall be bright, well sanded, and free from brashness, discolorations, worm holes, splits, and shake and honeycomb. Wood for unexposed parts shall be free of brashness and honeycomb. Wood parts shall be uniformly kiln dried to a moisture content of from 6 to 10 percent and shall be stored in properly controlled atmosphere to assure uniform distribution of moisture among the individual pieces.

3.2.2.2 Plywood. Plywood shall be Douglas Fir plywood, or other hardwood plywood, veneer core construction conforming to the requirements of Commercial Standard CS35.

3.2.3 Filler. The filler when used, shall be compatible with the finish system and shall not cause failure to finish pannels in tests specified in 4.3.1.1.

3.2.4 Stain. The stain shall consist of an organic dye in an organic solvent and shall not cause failure to the finish panels in ultraviolet resistance test specified in 4.3.1.1.

3.2.5 Wood finish. Wood finish shall consist of an organic coating system which when applied and dried will pass the tests specified 4.3.1.1.

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3.2.6 Tacks and staples. Tacks and staples shall be of commercial upholstery type and of blued steel. Tacks for securing strip webbing shall be 12 ounce size, those for securing cambric and welting shall be 3 ounce size and those for securing cover material shall be 6 ounce size. Staples for securing the webbing shall be not less than 0.050 by 0.020 inch wire size with legs 1/2 inch (minimum) long. Staples for securing burlap, cover cambric and welting material shall be of wire size not less than 0.030 by 0.020 inch with legs 3/8 inch long.

3.2.7 Glides. Glides shall be of the rubber cushioned nail type. The lower face (in contact with floor) of the glides as well as all exposed surface below the rubber cushion shall be highly polished stainless steel and shall be not less than 1 inch in diameter. The rubber cushion shall be a minimum of 1/8 inch thick. Shank shall be 7/8 inch long (min.), 0.050 inch thick and serrated.

3.2.8 Casters. Casters shall conform to type III - ball-bearing swivel, wood chair, material 1 - hard-rubber composition wheel with soft rubber tread (nonmarking) of FF-C-82. For type II, style A chairs, the casters shall be size A (1-5/8 inch or 1-3/4 inch diameter wheel), and for type II style B chairs, the casters shall be size B (2-inch diameter wheel). All casters shall have an oxidized copper finish.

3.2.9 Rotary mechanism. The rotary mechanism used for Type II chairs shall be metal having four contact points to the underside of the seat, a spiral spring tension adjustment or a rubber torsion unit adjustment for back tilt, and a positive stop to prevent the seat from going over backward should there be a failure of the tilt mechanism, and an adjustment for height of seat over a range of not less than 3 inches. Adjustments for tilting and raising shall be by hand wheels conveniently located.

3.2.10 Scuff plates. The scuff plates used on the bases of Type II chairs shall be dark brown molded phenolic resin compound.

3.2.11 Upholstering materials.

3.2.11.1 Fabric cover. The fabric shall be woven from a yarn composed of pigmented vinyl resin with a nylon core. It shall weigh 16 ounces per square yard minimum and shall have the following properties:

Thread count (minimum)	warp: 36	filling: 18
Tensile strength (grab) lbs. minimum	warp: 300	filling: 200
Tearing strength (trapezoid) lbs. average	warp: 70	filling: 35
Elongation (27 lb. weight), percent, maximum	warp: 5	filling: 4
Colorfastness	good	
Resistance to crocking	excellent	

3.2.11.2 Polyurethane. Polyurethane shall conform to Type II of Interim Federal Specification L-P-00386(GSA-FSS), Class 3 for seats and Class 2 for backs, except that the density shall be not less than 1.8 pounds per cubic foot.

3.2.11.3 Rubber webbing. Rubber webbing shall be of the width specified herein, and shall have a breaking strength of 300 pounds (minimum) and an elongation of 35 percent when tested as specified in Method 5102 of Federal Test Method Standard 191.

3.2.11.4 Cambric. The cambric shall be a plain weave cotton fabric weighing 3.2 ounces per linear yard on a 30-inch width basis and having a breaking strength of not less than 30 pounds in each direction. As an alternate material, a spun bonded polypropylene fibrous sheet weighing 2.0 ounces per square yard may be used.

3.2.11.5 Polyester filling materials.

3.2.11.5.1 Polyester fiberfill. Polyester fiberfill shall be composed of filament polyester fiber of 25-40 denier, lightly crimped. It shall weigh approximately 2 ounces per square foot.

3.2.11.5.2 Polyester fibercoil. Polyester fibercoil shall be composed of filament polyester fiber processed to yield a coil form. It shall weigh approximately 1.5 ounces per square foot.

3.3 Construction.

3.3.1 Design. The design shall approximate that shown in Figures 1 and 2.

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3.3.2 Dimensions. Overall finished dimensions shall be as shown in Table III.

Table III. Overall dimensions

	Type I		Type II	
	Style A	Style B	Style B	Style A
Height	30-1/2 - 31-1/2	30-1/2 - 31-1/2	30-1/2 - 31-1/2	30-1/2 - 31-1/2
Width	16 - 17	22 - 23	22-1/2 - 23-1/2*	17 - 18*
Depth	20	21 - 22	-	-
Effective seat depth	17	17 - 19	17 - 19	14 - 15
Seat width	16 - 17	19 - 20	19 - 20	17 - 18
Seat height	17 - 18	17 - 18	17**	17**

* Without base
** At lowest setting
*** Effective seat depth is defined as the distance from the front of the seat to the point on the seat which would be intersected by projection of the front face of the back.

All single number dimensions are minimum.
The height of the chair backs shall be 9 - 10 inches.

3.3.3 Joinery. All frames shall be rigidly constructed of the hardwoods specified in 3.2.2 and assembled as shown in the illustrations and as specified herein. Posts shall be made from 6/4 stock. Front, side and back rails shall be made from 5/4 stock. Built-up stock will be permitted provided samples of the same lay-up pass the test in 4.4.2. All joints shall be fitted carefully, thoroughly glued and secured with not less than two dowels or screws and bolts. Mortise and tenon joints may be used in lieu of dowels provided the tenons are at least three-fourths of the width of the piece and at least 3/4 inch deep. Corner blocks not less than one inch thick and 1/16 to 1/8 inch less in height than the rails to which attached, shall be fitted carefully, glued and secured by not less than two screws in each side. All exposed squared edges shall be eased. The height of the seats at the front posts shall be approximately 1-1/2 inches greater than the back posts. The back pitch (from vertical) shall be 8 - 10 degrees.

3.3.4 Type I, straight chairs. Type I chairs shall be either with arms, Style B, Figure 1 or without arms, Style A, Figure 1.

3.3.4.1 Frame. The frame shall be constructed as specified in 3.3.3. Frame component dimensions shall be as shown in Table IV. The front and back seat rails shall be curved.

Table IV. Frame component dimensions

Component	dimensions (inches), minimum
Arms, Style B only	1-1/4 by 1-1/2
Front and back posts	1-1/4 by 1-1/2
Seat rails	1 by 2-1/2
Back rest rail	1 by 1-3/4
Seat front rail (curved)	
Back panel (Type II, Style A)	3/8 by 9
Seat panel	1/2
Back assembly (other than Type II, Style A)	
Top rail	1-1/4 by 2
Bottom rail	1-1/4 by 2-1/2
Sides	1-1/4 by 1-1/4
Base reinforcement (Type I, Style A)	
Side stretchers	3/4 by 3/4
Cross stretcher	3/4 by 1

3.3.4.2 The back shall be completely upholstered without welts. It shall consist of a frame with top and bottom rails joined by vertical side pieces and the assembly doweled to the back post using two dowels (7/16 inch diameter, minimum) at each joint. The dimensions of the frame members shall be as specified in table IV. There shall be a piece of 16 ounce (minimum) sheet burlap or 3.5 ounce spun bonded polypropylene sheet supplied over the frame, doubled at the sides and secured with tacks or staples. A layer of polyurethane as specified in 3.2.11.2, one inch thick shall be applied over the inside face of the base fabric, followed by the vinyl fabric specified in 3.2.11.1. No seam shall be visible on the front, back or top surface of the back.

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3.3.4.3 Seat. The seat shall be constructed by either of the following methods:

a. The seat shall consist of 7-ply hardwood plywood, 1/2 inch thick with 5 to 7 holes drilled through it. The edges shall be eased, set in a frame and supported additionally by corner blocks. A thin layer of polyester Fibercoil shall be applied over the plywood pan, followed by a 2-inch layer of polyurethane and a layer of polyester Fiberfill batting sewn between cheesecloth and then the final upholstery fabric cover.

b. Alternatively, the seat shall consist of a rubber webbing platform formed with five 1-1/2 inch or four 2-inch (minimum) wide strips of rubber webbing from front to back and four 1-1/2 or three 2-inch wide interlaced strips from side to side. The webbing strips shall be evenly spaced and fastened securely to the seat frame. A 2-inch polyurethane layer as specified in 3.2.11.2 plus a layer of polyester Fiberfill shall be applied and covered with the upholstery cover of the color specified.

3.3.5 Type II, rotary chairs. Type II chairs shall be either with arms, Style B, Figure 2, or without arms, Style A, Figure 2.

3.3.5.1 Frame, Style B. The frame shall be the same as that specified in 3.3.4.1.

3.3.5.2 Back, Style B. The back shall be the same as that specified in 3.3.4.2.

3.3.5.3 Seat, Style B. The seat shall be the same as that specified in 3.3.4.3 except that the polyurethane layer shall be 3 inches thick for both methods a and b. The seat shall be attached to the spider arms of the rotary mechanism specified in 3.2.9 by four 1/4-inch bolts.

3.3.5.4 Base, Style A. The base shall be as shown in Figure 1 and shall consist of four legs which shall be received into a steel hub by means of a continuous dovetail extending the full length of the hub. The dovetail shall be tapered in design so that when the dovetails of the wooden legs are inserted into the dovetails of the hub, pressure will be required to complete the joining. Such base assemblies shall be reinforced on the bottom by a rust-resistant steel or malleable iron butterfly plate which shall be secured to the legs by means of eight 1-1/2 inch, No. 12 screws, two to each leg. Part rounded wood filler blocks shall be recessed into adjoining legs to conceal the hub and shall be glued on one side only. The base assembly shall be free of play when moved in any direction. A scuff plate, as specified in 3.2.10, shall be mounted on each leg. The span of the legs shall approximate the diagonal distance across the seat.

3.3.5.5 Seat, Style A. The seat shall be constructed by either of the following methods:

a. The seat shall consist of 7-ply hardwood plywood 1/2 inch thick, curved and having 5 to 7 holes drilled through it. The edges shall be eased, set in a frame and supported additionally by corner blocks. A thin layer of self-adhering polyester Fibercoil shall be placed over the plywood pan, followed by a 2-inch layer of polyurethane and a layer of polyester Fiberfill batting sewn between cheesecloth and then the final upholstery cover.

b. Alternatively, the seat shall consist of a rubber webbing platform formed with four 1-1/2 inch or three 2-inch wide strips of rubber webbing from front to back and three 1-1/2 or two 2-inch wide interlaced strips from side to side. The webbing strips shall be evenly spaced and fastened securely to the seat frame. The front and back rails shall be curved. A 2-inch polyurethane layer plus a layer of polyester Fiberfill shall be applied and covered with the upholstery cover of the color specified.

3.3.5.6 Back. The back shall consist of a 7-ply hardwood panel, 1/2 inch thick, curved and with edges grooved. The panel shall be completely upholstered using one inch thick polyurethane as specified in 3.2.11.2 firmly adhered to the wood and followed by the final upholstery cover. The finished back shall be nine inches high and 15 inches wide.

3.3.5.7 Back assembly. The completed back shall be fastened to two tapered hardwood posts made from one piece or built-upstock. Post shall extend from the backplate of the rotary mechanism to within one inch of the top of the back. The post shall be 7/8 inch minimum thick, 2-1/2 inches wide at the seat tapering uniformly to 2 inch width at the top. The posts shall be secured to the backplate of the rotary mechanism by bolts and screws. Posts shall be secured to the back by two countersunk screws in each post. Each exposed screw head shall be finished in inconspicuous brown lacquer. There shall be a cross back rail between the posts, located so that its top surface is approximately 1/4 inch below top of posts. Rail shall be 7/8 inch (minimum) thickness and 2-1/4 inches wide, and shall be doweled to posts.

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3.4.1 Chromium plating. Chromium plating shall conform to Class 1, Type II of QQ

3.3.5.8 Base. The base shall be as specified in 3.3.5.4 and shall follow the contours shown in Figure.

3.3.6 Upholstery. Covers shall be tailored carefully and so applied that they are neat and tight, free from wrinkles or bulges. The kind and size of thread as well as the type and number of stitches shall be suitable for the kind of fabric and the service intended. Seam allowances shall be maintained and seams sewn properly so that no raw edges, runoffs, broken or skipped stitches, twists, pleats, or puckers result. Covers for chair seats shall be tacked or stapled using tacks or staples specified in 3.2.6. The underside of all chair seats shall be covered with cambric or the alternate fabric specified in 3.2.11.4.

3.3.7 Finish.

3.3.7.1 Wood surfaces. All wood surfaces normally exposed to view shall be finished sanded smooth and all corners and edges eased, thoroughly cleaned, and finished using an organic coating system, the dried film of which shall pass the tests in 4.3.1.1.

3.4 Color and appearance of finish.

3.4.1 Exposed wood. The color and appearance of the finish of exposed wood surfaces shall be walnut and shall conform to the standard reference panel adopted by Federal Supply Service, General Services Administration for color, texture and gloss (see 6.3).

3.4.2 Upholstery fabric. The color of the upholstery fabric shall be bittersweet, FSS-06-S-F, saffron, FSS-26-S-F, peacock, FSS-36-S-F, and black, FSS-85-S-F, as specified (see 6.2). The color, finish, gloss, texture, and appearance shall conform to the standard samples adopted by Federal Supply Service, General Services Administration (see 6.3).

3.5 Identification marking. Each item shall be permanently and legibly marked in an inconspicuous place with the specification number, Federal Stock Number, contract number, date of manufacture and the manufacturer's name or trade-mark, so that the source of supply may be readily determined.

3.6 Workmanship. Chairs covered by this specification shall be produced using methods of construction, assembly, upholstering, and finishing which will result in sturdy seating pieces with good appearance. The occurrence of defects shall not exceed the acceptable quality levels specified herein. Whenever this specification does not prescribe the component or process in detail, good commercial practice in the furniture industry shall prevail.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.2 Government verification. All quality assurance operations per formed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of surveillance of the operations to determine whether the practice, methods, and procedures of the written inspection plan are being properly applied, and Government product inspection, to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices that might have an effect on the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct the deficiencies discovered shall be cause for suspension of acceptance until the correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances.

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4.2 Preproduction sample inspection. Examination and testing shall be made on the completely fabricated item and the cut-away sample for all provisions of this specification.

4.3 Sampling for inspection and acceptance. Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.3.1.1 Testing of components. In addition to the quality assurance testing provisions of the subsidiary specifications, testing of the components shall be performed in accordance with Table V for the characteristics shown therein. The values specified in section 3 of this specification apply to the average determinations made on a sample unit for test purposes as specified in the applicable test method. When the sample size for components covered by the subsidiary specifications is not specified, the sample size for each characteristic shall be in accordance with inspection level S-2 of MIL-STD-105 except that for the lot sizes of 501 (units of products) and over the sample size shall be 5. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified.

Table V. Instructions for testing components and materials

Component	Characteristic	Requirement Reference	Test Method	Sample Unit
Adhesive	Block shear test	3.2.1	ASTM D 905 1/ 2/ 3/	1/2 pint
Wood Finish System (Stain, sealer, filler, and final coating 4/	Spot test - alcohol	3.2.3, 3.2.4, and 3.2.5	4.3.3.3 and 4.3.3.3.1 of TT-L-57	1/2 pint of each
	Print resistance	3.2.3, 3.2.4, and 3.2.5	4.3.3.4, 4.3.3.4.1, 4.3.3.4.2 and 4.3.3.4.3 of TT-L-57	
	Cold check resistance	3.2.3, 3.2.4, and 3.2.5	4.3.3.5, 4.3.3.5.1, 4.3.3.5.2, and 4.3.3.5.3 of TT-L-57	
	Boiling water	3.2.3, 3.2.4, and 3.2.5	4.3.3.6 of TT-L-57	
	Ultra-Violet resistance	3.2.3, 3.2.4, and 3.2.5	4.3.3.7 of TT-L-57	
	Toughness and adhesion	3.2.3, 3.2.4, and 3.2.5	4.4.1 herein	
Rubber webbing	Breaking strength and elongation	3.2.11.3	Method 5102 of Fed. Test Method Std. No. 191	2 linear yards
Edge-glued joint	Joint integrity	3.2.2	4.2.2	one joint
Upholstery fabric	Thread count		Method No. 5050	1 square yard
	Tensile strength		Method No. 5100	
	Tearing strength	3.2.11.1	Method No. 5136	
	Elongation		Method No. 5100	
	Resistance to crocking		Method No. 5651 Fed. Test Method Std. No. 191	
Cambric	Weight	3.2.11.4	Method No. 5041 of Fed. Test Method Std. No. 191	1/4 yard
	Breaking strength		Method No. 5100	
Cambric alternate	Material identification	3.2.11.4	Certificate	1/4 yard
	Weight		Method No. 5041 of Fed. Test Method Std. No. 191	

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Table V. Instructions for testing components and materials (continued)

Component	Characteristic	Requirement Reference	Test Method	Sample Unit
Polyester fiber	Material identification Weight	3.2.11.5.1	Certificate Method No. 5041 of Fed. Test Method Std. No. 191	1/2 yard
Polyester fibercoil	Material identification Weight	3.2.11.5.2	Certificate Method No. 5041 of Fed. Test Method Std. No. 191	1/2 yard

1/ The following modifications apply when testing casein-type adhesive. The test blocks shall be placed under pressure between 10 and 20 minutes after spreading the adhesive, the adhering pressure shall be 175 ± 25 pounds per square inch, and the test blocks shall be held under pressure for at least four hours.

2/ The following modifications apply when testing animal adhesive. The moisture content of the wood shall be from 6 to 9 percent. The joints shall be conditioned at a relative humidity of 40 ± 4 percent.

3/ Whenever a specimen fails at a load of less than 2,800 pounds per square inch and the failure occurs 50 percent or more in the wood, the specimen shall be disregarded in computing the average. In case the shearing strength of one or more specimens is 2,800 pounds per square inch or higher, the variation among individual specimens is 10 percent or more (based on the strongest joint) the test shall be run again, provided the average is less than the required 2,800 pounds per square inch.

4/ The test shall be made on completely finished wood panels of the same wood used in production. The finish shall be in accordance with procedures used in producing furniture.

4.3.2 Intermediate inspection.

4.3.2.1 Visual examination of the intermediate product. The lot size shall be expressed in units of fabricated chairs prior to application of the upholstery cover. The inspection level for purposes of determining the sample size shall be level II of MIL-STD-105. The unit of product for these examinations shall be one fabricated chair prior to application of the upholstery cover. The defects found during the examinations shall be classified in accordance with Table VI. The acceptable quality level (AQL) shall be 15.0 for major defects and 40.0 for total defects (major and minor combined) per hundred units.

Table VI. Intermediate product defects

Examine	Defect	Classification Major	Classification Minor
Framing members	Species of wood in any wood component not as specified	X	
	Decay over 1/4 square inch in area in any component	X	
	Decay up to and including 1/4 square inch in area in more than one component		X
	Bark pocket over 1 inch long		X
	Bark or wane over 1 square inch in area		X
	Split, shake, or check over 2 inches in length	X	
	Split, shake, or check up to and including 2 inches in length which may affect serviceability		X
	Loosened grain, torn grain or chip-out which may affect serviceability		X
	Area over 4 square inches not smoothly surfaced		X
	Honeycomb	X	
	Decayed or loose knot over 1 inch average diameter	X	

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Table VI. Intermediate product defects (continued)

Examine	Defect	Classification Major	Classification Minor
Framing members (con't)	Decayed or loose knot up to and including 1 inch average diameter		X
	Sound tight knot on edge of piece of wood or at a construction point.	X	
	Sound tight knot over 3/4 inch average diameter		X
	More than two sound tight knots in one piece of wood		X
	Sapwood or mineral streak which may affect serviceability or strength		X
	More than two wormholes in component that will be unexposed after upholstery attachment		X
	Undesired wood defect filled in with plastic wood or other filling compound		X
	Plywood not of construction specified	X	
Workmanship of joinery	Wood component missing	X	
	Loose or unsound joint	X	
	Evidence that glue at joint or block has not been applied where required		X
	Frame not constructed as specified	X	
	Evidence that joint is not type specified	X	
	Frame out of square, i.e., sprung	X	
	Two or more screws missing	X	
	Screw missing where required		X
	More than one screw loose or not driven in all the way		X
	Screw point protrudes from the wood surface .		X
	Corner block or glue block not accurately fitted into place		X
	Wood member improperly positioned or not positioned as specified	X	
	Wood member warped, cupped, or twisted out of specified shape	X	
	Shape or contour of wood member not as specified	X	
	Wood not full length piece where required		X
	Component not dimensions specified (including tolerance)	X	

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Table VI. Intermediate product defects (continued)

Examine	Defect	Classification Major	Minor
Construction and Workmanship of Platform	Webbing strips not number specified	X	
	Webbing strips not interwoven as specified		X
	Webbing strips not secured to seat frame as specified	X	
	Webbing strips not equally spaced.	X	
Polyurethane Cushion			
Workmanship	Internal voids perceptible to sight or touch through uncut surfaces	X	
	Presence of unrepaired surface voids 1/8 inch or greater in depth and 1/2 inch or larger in width on top or bottom	X	
Vertical surfaces	Unrepaired exposed voids greater than one inch in width and exceeding 1/4 inch in depth	X	
Horizontal cut surfaces	Unrepaired voids greater than 3/4 inch in width and more than 1/8 inch in depth or greater than 1/2 inch in width and more than 1/4 inch in depth	X	
Upholstery fabric	Color of material not as specified	X	
	Torn, cut, or contains hole	X	
	Misweave, i.e., multiple float or other fabric damage affecting serviceability		X
	Spotted, stained or dirty		X
	Shade variations or faded, affecting appearance	X	
	Material has objectionable odor		X
Construction and Workmanship of Upholstery	Material pieced where a continuous piece of material is required	X	
	Seam open up to and including 2 inches in length		X
	Seam open over 2 inches in length	X	
	Seam uneven or not neatly accomplished affecting appearance		X
	Seam in wrong location		X
	Needle chews		X
	Skipped stitches or runoffs		X
	Raw edge noticeable at seam		X
	Color of thread does not match color of upholstery material		X

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4.3.3 End item inspection.

4.3.3.1 Visual examination of the end item. The lot size shall be expressed in units of completed chairs. The inspection level for purposes of determining the sample size shall be level I of MIL-STD-105. The unit product shall be one completed chair. The defects found during the examinations shall be classified in accordance with Table VII. The acceptable quality level (AQL) shall be 10.0 for major defects and 25.0 for total defects (major and minor combined) per hundred units.

Table VII. End item defects

Examine	Defect	Classification Major	Classification Minor
Construction and Workmanship of all Chairs	General design or styling not as specified or not as on approved preproduction sample	X	
	Contour of seat or back not in accordance with figure and not as specified	X	
	Legs uneven (Type I); item rocks more than 1/16 inch on flat surface.	X	
	Component missing	X	
	Cambric if used, not applied with glazed side exposed		X
	Tucking of material at corner or fold not neatly accomplished affecting appearance		X
	Cover wrinkled or puckered at seam to an extent adversely affecting appearance		X
	Cambric or alternate fabric not neatly attached		X
	Furrow, lump or bulge affecting appearance		X
	Component in wrong location		X
	Component not size and shape specified	X	
	Component not assembled as specified	X	
	Component out of alignment or out of square affecting serviceability	X	
	Component loose or insecurely fastened where rigidity is required	X	
	Poor fit of component part or to its mating part	X	
	Any tack or other protuberance that can be felt through the upholstery		X
	Evidence of filler voids		X
Construction and Workmanship of metal parts	Weld or braze missing where required	X	
	Weld or braze broken, not fused or burned through	X	
	Weld or braze incomplete where continuity is required or undercut	X	
	Flash or spatter not removed	X	
	Weld or braze not ground smooth or not neatly accomplished	X	
	Contains hole where not required, does not have one where required		X

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TABLE VII. End item defects (con.)

Examine	Defect	Classification Major	Minor
Contruction and Workmanship of metal parts	More than one screw missing	X	
	Screw or bolts broken off		X
	Fin, burr or sharp edge	X	
	Glide missing (Type I chairs)		X
	Glide not type or material specified (Type I chairs)		X
	Glide wrong size (Type I chairs)		X
	Casters not as specified (Type II chairs)	X	

4.3.4 Dimensional examination. The lot size shall be expressed in units of chairs. The inspection level shall be S-2 of MIL-STD-105. The unit of product shall be one chair. Any external overall dimensions that are not within the specified tolerance shall be classified as a defect. The acceptable quality level (AQL) shall be 4.0 defects per hundred units.

4.3.5 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking, as required by section 5 are complied with. Defects shall be as indicated in table VIII. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE VIII. Visual defects, preparation for delivery

Examination	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closue of containers. Cushioning inadequate or missing. Bulged or distorted container.
Contents	Content per container is more or less than required.

4.3.6 Examination for closure and strapping of shipping containers. Examination for defects in closure and strapping of shipping containers shall be in accordance with appendix to PPP-B-636.

4.4 Test methods. The following test methods shall be used where referenced in this specification:

4.4.1 Toughness and adhesion test. The toughness and adhesion test shall be made by using Bell Laboratories Adhesion Tester, U. S. Testing Company, Model Number 1001, in accordance with instructions for using, as provided for by the manufacturer of the tester (see 6.4). The panel shall be marred both parallel and perpendicular to the grain. The resulting indentation shall be examined for whitening or cracking.

4.4.2 Edge-glued joint test. A chisel test on edge-glued joints shall be conducted as follows:

a. Place a sample component with an edge-glued joint end wise on a bench or other solid object.

b. Take a chisel, wider than the thickness of the member being tested, and carefully align it on the glue joint on the upper end of the member.

c. With the chisel being held firmly in a vertical position, strike it sharply with a mallet or hammer so as to split the member longitudinally. Care should be exercised to insure that the force of the blow is directed to the glue line. The resulting split should reveal a failure of the glue line or in the area adjacent to it.

d. Inspect the glue joint area to determine the kind of failure and estimate the amount of wood failure.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each chair shall have all finished surfaces of the front legs, back posts, rails, arm rests (when applicable) and top of back rest covered with a single sheet of 30 pound basis weight (24 x 36 - 500) kraft paper, waxed paper or polyethylene film, a minimum of 1.5 mils thick in accordance with L-P-378. These same surfaces shall be further protected with a single 3/8 inch thickness of cushioning material fabricated in the form of pads or blankets and may be any one of the following materials: Macerated paper, plain; macerated paper, tissue faced; shredded paper; wood wool; wood fiber felt; cellulosic cushioning material excelsior (sealed ends); cotton wadding or cellulosic cushioning material laminated to corrugated fiberboard. When cellulosic cushioning material, cotton wadding or tissue faced macerated paper are used, the initial wrap of kraft paper, waxed paper or polyethylene film may be omitted. The cushioning shall be secured in place with tape conforming to PPP-T-45, or twine having a minimum breaking strength of 30 pounds.

5.1.1.1 Casters. The casters, when applicable, shall be removed from the chair and placed in a single bag. The bag shall be a cotton drawstring type or heat sealed polyethylene type, and shall be secured to the rotary base assembly in a manner that will preclude movement or damage during shipment and storage.

5.1.2 Level B. The chairs shall be packaged as specified in 5.1.1 and 5.1.1.1.

5.1.3 Level C. The chairs shall be packaged in accordance with the manufacturer's standard practice, providing that this will insure protection for the item during shipment and provide for safe delivery to its destination.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I, style A chairs. Type I, style A chairs packaged as specified in 5.1 shall be packed two to a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant, variety DW, grade V15c. Close, waterproof seal and strap the containers in accordance with method II of the appendix to PPP-B-636.

5.2.1.2 Type I, style B chairs. Type I, style B chairs packaged as specified in 5.1 shall be individually packed in close-fitting boxes conforming to PPP-B-636, type CF, class weather resistant, variety DW, grade V15c. Close, waterproof seal and strap the containers in accordance with method II of the appendix to PPP-B-636.

5.2.1.3 Type II, style A chairs. Type II, style A chairs packaged as specified in 5.1 shall be prepared for packing by setting and securing the assembled chair in the lowest rotary position, or the seat and back assembly shall be removed from the rotary base assembly, thereby reducing the cubage to a practicable minimum. The chairs shall then be packed two to a close-fitting box conforming to PPP-B-636, class weather resistant, variety DW, grade V15c. Close, waterproof seal and strap the containers in accordance with method II of the appendix to PPP-B-636.

5.2.1.4 Type II, style B chairs. Type II, style B chairs packaged as specified in 5.1 shall be prepared for packing by setting and securing the assembled chair in the lowest rotary position, or the upper assembly shall be removed from the rotary base assembly, thereby reducing the cubage to a practicable minimum. The chairs shall then be individually packed in close-fitting boxes conforming to PPP-B-636, class weather-resistant, variety DW, grade V15c. Close, waterproof seal and strap the containers in accordance with method II of the appendix to PPP-B-636.

5.2.2 Level B. Chairs packaged as specified in 5.1 shall be packed as specified in 5.2.1.1 thru 5.2.1.4, as applicable, except that the boxes shall be in accordance with PPP-B-636, type CF, class domestic, variety DW, grade 200.

5.2.3 Level C. Chairs packaged as specified in 5.1.3 shall be packed in containers that will assure carrier acceptance and safe arrival at destination in compliance with the Uniform Freight Classification rules and the National Motor Freight Classification rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to the special markings required by 5.3.3, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military agencies. In addition to the special markings required by 5.3.3, the shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special Marking. In addition to the markings required by 5.3.1 or 5.3.2, the front and back panels of each shipping container shall be marked in each upper corner with the word "UP" and an arrow pointing to the direction in which the container should be shipped and stored. "UP" shall be in characters a minimum of 2 inches high. Additionally the top and one side and one end of the container shall be marked as follows in characters a minimum of two inches high. Top of container:

TOP
THIS SIDE UP
FURNITURE, FRAGILE-HANDLE WITH CARE

One side and one end of container:

FURNITURE
FRAGILE-HANDLE WITH CARE

6. NOTES

6.1 Intended use. Chairs covered by this specification are intended for use in general office areas with desks, tables, bookcases, and other units to form work stations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

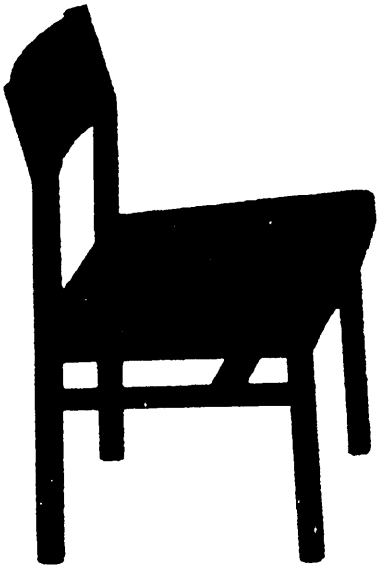
(a) Title, number, and date of this specification.
(b) Type and style, required if applicable (see 1.2.1).
(c) When preproduction samples not required (see 3.1).
(d) Color of fabric upholstery (see 3.3.6.1).
(e) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(f) Marking requirements (see 5.3).

6.3 Standard sample. Sample panels of the standard color for the walnut finish and upholstery fabric are obtainable, without charge, from the General Services Administration, Federal Supply Service, Washington, D. C. 20406.

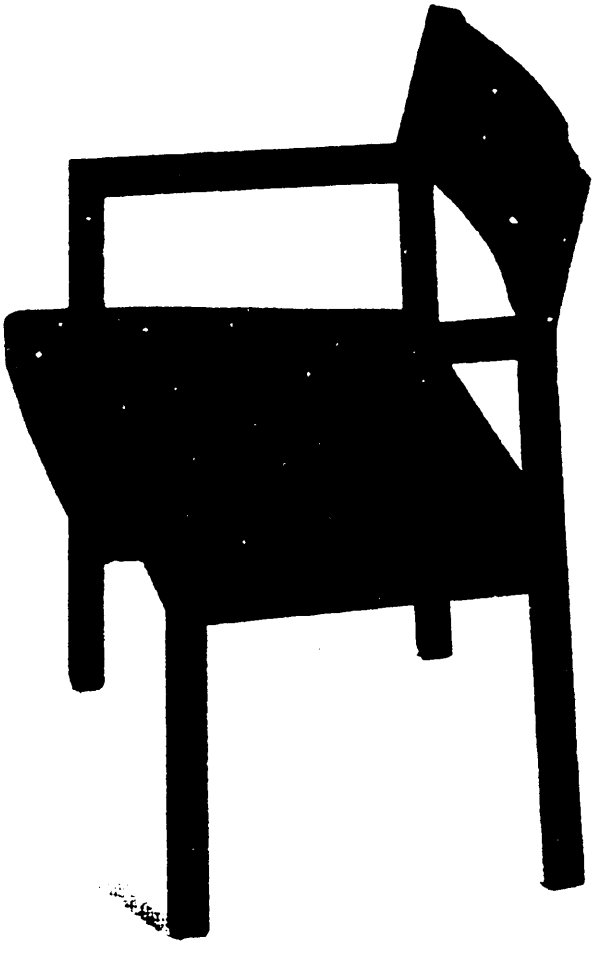
6.4 Information with respect to Bell Laboratories Adhesion Tester, U. S. Testing Company, Model No. 1001, may be obtained from U. S. Testing Company, Inc., 1415 Park Avenue, Hoboken, New Jersey.

GSA DC 72-3784

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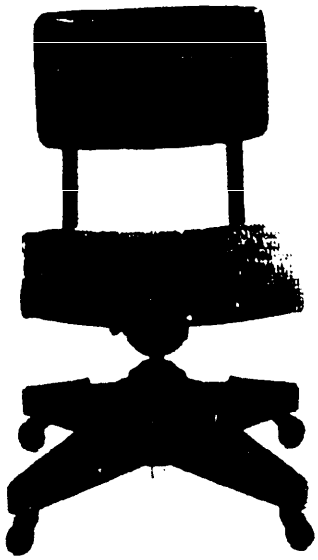
Style A



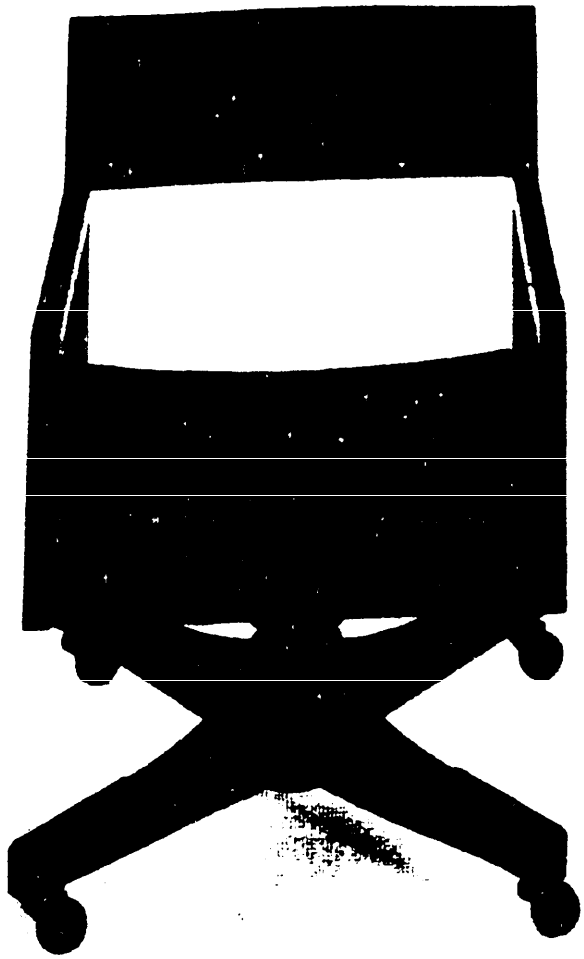
Style B

Figure 1. Type I, General Office Wood Chairs

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Style A



Style B

Figure 2. Type II, General Office Wood Chairs

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and **mail to: General Services Administration, FSS (FMSF), Washington, D. C. 20406.**

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION **AA-0-001628(GSA-FSS)**
CHAIRS, WOOD, GENERAL OFFICE

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT. AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

FORM
GSA FEB 69 2200

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